

MAT 310, Handout 24: Method of Distribution Functions (6.3)

Motivation: Given the distribution of the RV Y , we want to find the distribution of the RV $X = g(Y)$.

Method of Distribution Functions: Given the distribution of RV Y , we find the distribution of $X = g(Y)$ by finding $F_X(x) = P(g(Y) \leq x)$.

Example 1. The density of Y is given by

$$f_Y(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1 \\ 0, & o.w. \end{cases}.$$

Find the density function of $X = 2Y - 1$.

Example 2. The density of Y is given by

$$f_Y(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1 \\ 0, & o.w. \end{cases}.$$

Find the density function of $W = 4Y^2 + 1$.

Example 3. Suppose Y_1 and Y_2 are independent $U(0, 2)$ RVs. Find the density function of $X = Y_2 - Y_1$.

